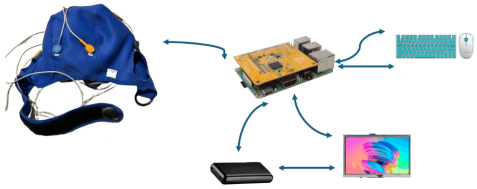
QUICK START & SAFETY GUIDE · SKU CQA240229P1

From a Bare Pi to Live EEG

Seven steps from a Raspberry Pi to live EEG in the browser: prepare the Pi, mount the shield, connect the electrodes, power on from the battery, install PiEEG Server and check the signal.



WARNING Battery only.

PiEEG-8 has **no isolation circuitry**. The Raspberry Pi, the shield and everything cabled to them (monitor, keyboard, mouse) must run from a **5 V power bank** while electrodes are on a person. PiEEG-8 is **not a medical device**.

01 SAFETY RULES — READ BEFORE PUTTING ELECTRODES ON ANYONE

#	Rule
1	**Battery only.** Pi, shield and everything cabled to them run from a 5 V power bank.
2	**No mains connection of any kind:** no wall charger, no desk supply, no USB or HDMI cable to a mains-powered computer, monitor or TV.
3	**No diagnosis.** Do not use PiEEG-8 to diagnose, monitor or treat any medical condition.
4	**Implanted devices.** Not on a person with a pacemaker or other implant without a physician's approval.
5	**Skin.** Clean, intact skin only. Stop if it becomes irritated; clean off gel after each session.
6	**Exposed electronics.** No case: handle by the edges, use anti-static precautions, keep liquids away.

Power while recording

Setup	While electrodes are on	Why
Pi + shield on a 5 V power bank	SAFE	No path to the mains.
Monitor powered by the power bank or the Pi	SAFE	Still fully on battery.
Laptop on battery, dashboard over Wi-Fi	SAFE	No cable to the Pi or the person.
Pi on a wall charger	NEVER	Connects the person to the mains through the electrodes.
Monitor plugged into a wall socket	NEVER	The HDMI cable links the mains to the Pi.
USB / Ethernet to a mains-powered computer	NEVER	Same reason.

02 WHAT YOU NEED

Item	Requirement
PiEEG-8	Included
Raspberry Pi	Pi 4 or Pi 5, microSD 16 GB+ with Raspberry Pi OS (64-bit, desktop)
5 V power bank	5 V, 3–3.5 A, up to 10,000 mAh; short, thick USB cable
Electrodes	8 dry or wet Ag/AgCl with 2.54 mm connectors + 2 ear clips
EEG cap	10-20 nylon cap or DIY
Monitor, keyboard, mouse	Monitor powered from the bank or the Pi; wired keyboard and mouse
Gel	Only for wet electrodes

03 SETUP IN SEVEN STEPS

1 Gather the parts

PiEEG-8, Raspberry Pi 4 or 5, charged power bank, 8 electrodes and 2 ear clips, cap.

2 Prepare the Raspberry Pi

Flash Raspberry Pi OS (64-bit, with desktop) with Raspberry Pi Imager, boot once and connect to the internet for installing software. SPI is enabled by the PiEEG Server installer.

3 Mount the shield

Shut down the Pi and unplug the power bank. Line up pin 1 with pin 1 on the 40-pin header and press down evenly until every pin is seated.

4 Connect the electrodes

Channels 1–8 to the cap electrodes, **Ref** to one earlobe, **Biasout** to the other. See PE8-PIN-004.

5 Power on from the battery

Connect the battery-powered monitor and wired keyboard/mouse, then plug the power bank into the Pi. The LED on the shield lights up.

6 Install and start PiEEG Server

TERMINAL ON THE PI

```
curl -sSL https://raw.githubusercontent.com/pieeg-club/PiEEG-server/main/install.sh | bash
sudo reboot                      # first time only: enables SPI
sudo sed -i 's|bin/pieeg-server|bin/pieeg-server --device pieeg8|' \
     /etc/systemd/system/pieeg-server.service
sudo systemctl daemon-reload && sudo systemctl restart pieeg-server
```

The service starts in 16-channel mode by default; the `sed` line switches it to PiEEG-8. Check the setup with `pieeg-server doctor`.

7 Check the signal

Open **http://localhost:1617** on the Pi (or http://raspberrypi.local:1617 from a battery-powered laptop) and click **Connect**.

04 FIRST SIGNAL CHECKS

Test	What you should see
Sit still for 30 s	All 8 channels settle. Flat lines or huge values mean a loose electrode.
Close the eyes for 10 s	Alpha (8–13 Hz) rises on O1 and O2, then drops when the eyes open.
Blink a few times	Large slow spikes on Fp1 and Fp2.
Clench the jaw	Bursts of high-frequency noise — muscle (EMG), not brain activity.

05 TROUBLESHOOTING

Power and hardware

Problem	Fix
LED on the shield is off	Check the power bank is on and charged, and the shield is seated on all 40 pins.
Pi restarts / lightning bolt	Power bank too weak. Use 5 V / 3 A and a short, thick cable.
Shield off by one pin	Shut down, remove and re-seat; pin 1 must match pin 1.

SPI, GPIO and software

Problem	Fix
`/dev/spidev0.0` not found	SPI is off: sudo raspi-config → Interface Options → SPI → Enable, reboot.
Permission denied on SPI/GPIO	`sudo usermod -aG spi,gpio $USER`, then log in again.
Pi 5: gpiochip4 not found	Run `gpiodetect`, use the pinctrl-rp1 chip, or pass --gpio-chip to PiEEG Server.
Pi 5: RPi.GPIO error	`pip install rpi-lgpio` (drop-in for Pi 5).
externally-managed-environment	Use a Python virtual environment (PE8-SW-006).
Dashboard shows 16 channels	Start with --device pieeg8 and add it to the service.
Address already in use	Stop the service first: sudo systemctl stop pieeg-server.

Signal

Problem	Fix
All channels noisy at once	Check the **REF and BIAS ear clips** first.
One channel flat or very large	That electrode is not touching the skin.
Strong 50/60 Hz hum	Something is on the mains — unplug every charger, run the monitor from the battery.
Slow spikes on front channels	Blinks and eye movements.
Bursts of high-frequency noise	Jaw, face or neck muscles.
Alpha does not rise with eyes closed	Check O1/O2 touch the scalp; keep the eyes closed 10 s in a quiet room.

> **NOTE Board or setup?**
> Set all channels to input-shorted mode (`CHnSET = 0x01`). If the signal stays within a few µV, the hardware is fine and the noise comes from cables, electrodes or surroundings.

06 RELATED DOCUMENTS

PE8-DS-001 Product Datasheet	**PE8-MD-002** Mechanical & Dimensional Drawing	**PE8-PCB-003** PCB & Manufacturing Specification
PE8-PIN-004 Connector Pinout & Electrode Guide	**PE8-SW-006** Data Interface & Software Guide	**PE8-ENC-007** 3D-Printed Enclosure Guide